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26 August 1998
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3 May 1989

DETAIL SPECIFICATION

ALUMINUM ALLOY ARMOR, 2219, ROLLED PLATE AND DIE FORGED SHAPES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers aluminum alloy 2219 armor rolled plate and die forged shapes from 3/4 inch up to and including 4 inches in thickness.

1.2 Weldability. Alloy 2219 is weldable to itself and other alloys of the 2000 series, but not to alloys of 5083 type materials and aluminum-zinc-magnesium series by present welding techniques.

1.3 Classification. Material furnished under this specification should be one of the following types and tempers:

Rolled plate - T851
Die forged shapes - T852

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Research Laboratory, Weapons and Materials Research Directorate, ATTN: AMSRL-WM-M, Aberdeen Proving Ground, MD 21005-5069 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-662 - V_{50} Ballistic Test for Armor

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Automated Printing Service (DAPS), Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

SOCIETY OF AUTOMOTIVE ENGINEERS, Inc. (SAE)

AMS 2750 - Pyrometry (DoD adopted)
AMS 2772 - Heat Treatment of Aluminum Alloy Raw Materials

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI H35.2 - Dimensional Tolerances for Aluminum Mill Products (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Customer Service, 11 West 42nd Street, New York, NY 10036).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B557 - Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products (DoD adopted)
ASTM E34 - Chemical Analysis of Aluminum and Aluminum-Base Alloys (DoD adopted)
ASTM E227 - Optical Emission Spectrometric Analysis of Aluminum and Aluminum Alloys by the Point-to Plane-Technique (DoD adopted)
ASTM E607 - Optical Emission Spectrometric Analysis of Aluminum and Aluminum Alloys by the Point-to-Plane Technique, Nitrogen Atmosphere (DoD adopted)
ASTM E716 - Sampling Aluminum and Aluminum Alloys for Spectrochemical Analysis
ASTM E1251 - Optical Emission Spectrometric Analysis of Aluminum and Aluminum Alloys by the Argon Atmosphere, Point-to-Plane, Unipolar Self-Initiating Capacitor Discharge (DoD adopted)

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ASTM G47 - Determining Susceptibility to Stress-Corrosion Cracking of High Strength Aluminum Alloy Products (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2) and before production has commenced, samples of the specified item shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples and test forgings shall be manufactured by the process proposed for use on production.

3.2 Chemical composition. Chemical composition of material furnished under this specification shall be within the limits specified in table I.

TABLE I. Chemical composition, percent (ladle analysis).[1/]

Element	Min.	Max.
Silicon	-	0.20
Iron	-	0.30
Copper	5.8	6.8
Manganese	0.20	0.40
Magnesium	-	0.02
Zinc	-	0.10
Titanium	0.02	0.10
Zirconium	0.10	0.25
Vanadium	0.05	0.15
Others, each	-	0.05
Others, total	-	0.15
Aluminum, by difference		Remainder

[1/] See 6.5

3.3 Mechanical properties. Unless otherwise specified in the contract or order (see 6.2), the mechanical properties of rolled plate and forged shapes shall meet the requirements of table II, when tested as specified in 4.7.2. If mechanical properties differ from the values shown in table II, or if any other properties are required, the ballistic requirements shall be as agreed upon between the procuring activity and the manufacturer.

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TABLE II. Mechanical properties.

	Direction	Thickness, inches	Yield Tensile strength min. ksi (MPa)[1/]	strength 0.2% offset min. ksi (Mpa)[1/]	Elongation in 2 inches min. percent
Rolled plate	Long transverse	0.250-1.000	62.0 (427)	46.0 (317)	8
		1.001-2.000	62.0 (427)	46.0 (317)	7
		2.001-3.000	62.0 (427)	45.0 (310)	6
		3.001-4.000, incl.	60.0 (414)	44.0 (303)	5
Die forgings	Parallel to grain flow	Up to 4.0, incl.	60.0 (414)	48.0 (331)	7
	Transverse to grain flow	Up to 4.0, incl.	60.0 (414)	46.0 (317)	4

[1/] SI values for information only.

3.4 Ballistic requirements. Each ballistic test plate or test forging shall be proof fired for resistance to penetration by determining a ballistic limit protection, BL(P), using the applicable projectiles and obliquities shown in tables A-I, A-II, A-III and A-IV of Appendix A. The minimum requirements shall be based on the measured thickness and shall be as specified in these tables.

3.5 Stress-corrosion resistance. When stress corrosion testing is specified in the contract or order (see 6.2), plate 0.75 inch and greater in thickness shall be resistant to stress corrosion cracking, with 3 out of 3 specimens showing no evidence of cracking at the end of 10 days.

3.6 Thermal processing. Heat treatment shall conform to the requirements of AMS 2750 and AMS 2772 and shall be such as to enable the material to meet the requirements of these specifications.

3.7 Dimensions. (See 6.2.)

3.7.1 Rolled plate. Size of plate shall be as specified in the invitation for bids, contract or order (see 6.2).

3.7.2 Forgings. The forgings shall conform to the shape and dimensions on the drawings within any variations stated therein.

3.7.3 Tolerances, rolled plate. Unless otherwise specified in the contract or order (see 6.2), the plate submitted for production acceptance shall not vary from the specified dimensions, plus and minus, or from true straightness of edges by an amount greater than the tolerances shown for lengths, width, squareness, flatness and lateral bow as specified in ANSI H35.2.

3.7.3.1 Thickness. Tolerance in thickness of rolled plate material shall be as shown in table III.

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TABLE III. Thickness tolerances.

Ordered thickness, inches	Tolerance, inch Minus	Plus
0.750-0.875, incl.	0.030	1/
0.876-1.125, incl.	.035	"
1.126-1.375, incl.	.040	"
1.376-1.625, incl.	.045	"
1.626-1.875, incl.	.050	"
1.876-2.250, incl.	.055	"
2.251-2.750, incl.	.060	"
2.751-3.000, incl.	.065	"
3.001-4.000, incl.	.070	"

1/ The values for the plus column are derived from the full range of tolerances allowed in table 3.1 of ANSI H35.2 less the value shown in the minus column. For example, the tolerance on a 2.0 inch thick by a 73 inches wide plate would be as follows:

Thickness tolerance (from ANSI H35.2)	+ 0.080"
Full tolerance less minus tolerance	0.160 - 0.055 = 0.105"
Derived plus tolerance	0.105"
Tabulated minus tolerance	0.055"

3.8 Identification marking. Each plate or forging shall be marked on one surface or in at least one location with characters not less than 3/8 inch high with the following information. Impression stamping shall not be used, except in areas designated by the drawing.

(a) Manufacturer's name, trade mark, or both.
(b) Number of this specification.
(c) Alloy designation and temper.
(d) Ordered plate thickness in inches for rolled plate or Army part drawing number for forgings.

3.8.1 Ballistic test plates or forgings. In addition to the marking in 3.8, each ballistic test plate or forging shall be marked with the letters PRE for first article test plates or forgings and ACC for acceptance test plates or forgings. This marking shall be impression stamped on the edge of rolled plate and on the top and bottom of forged material in letters not less than 1/2 inch high. The intended impact area on forgings shall be clearly outlined.

3.9 Ballistic test plate or forging information. Unless otherwise specified in the contract or purchase order, a ballistic test report shall be submitted with each ballistic test plate or forging containing the information as specified in MIL-STD-662.

3.10 Workmanship.

3.10.1 Rolled plate. Rolled plate shall be uniform in quality and shall be clean, smooth and sufficiently free from buckles, blisters, hard spots, damaged ends, laminations and other defects which may affect its use in fabrication. Surface cracks, edge cracks or edge laminations shall be cause for rejection.

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3.10.2 Forged material. Forged material shall be uniform in quality and free from blisters, fins, seams, laps, cracks, laminations or other defects which may affect its use.

4. VERIFICATION

4.1 Clarification of inspections. The inspection requirements specified herein are classified as conformance inspection (see 4.2).

4.2 Conformance inspection. The test for acceptance of individual lots consists of testing all requirements specified in section 3. Verification of conformance inspection shall be in accordance with the provisions herein. Ballistic tests shall be performed at a Government test activity selected by the procuring activity (see 6.2).

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3.1).
(b) Conformance inspection (acceptance)(see 4.3.2).

4.3.1 First article inspection. First article inspection shall consist of the following:

(a) Chemical analysis (see 3.2 and 4.7.1).
(b) Mechanical properties (see 3.3 and 4.7.2).
(c) Ballistic tests (see 3.4, 4.7.4 and Appendix A).
(d) Stress corrosion tests (see 3.5 and 4.7.3.)
(e) Dimensions (see 3.7 and 4.8.1).
(f) Identification marking (see 3.8 and 4.8.2).
(g) Workmanship (see 3.10 and 4.8.2).

4.3.2 Conformance inspection (acceptance). Lot acceptance inspection shall consist of the following:

(a) Chemical analysis (see 3.2 and 4.7.1).
(b) Mechanical properties (see 3.3 and 4.7.2).
(c) Ballistic tests (see 3.4, 4.7.4 and Appendix A).
(d) Dimensions (see 3.7 and 4.8.1).
(e) Identification marking (see 3.8 and 4.8.2).
(f) Workmanship (see 3.10 and 4.8.2).
(g) Preparation for shipment (see section 4.8.3 and 5).

4.4 Lot. For rolled plate a lot shall consist of plate of the same ordered thickness and temper submitted for inspection as a unit, provided that the weight of the finished plate in the lot does not exceed 50,000 pounds. For forged material, a lot shall consist of forgings of the same shape, chemical composition limits and processing submitted for inspection as a unit, provided that the weight of the forgings in the lot does not exceed 25,000 pounds or 300 forgings, whichever is less.

4.5 Sampling.

4.5.1 Sampling for first article testing.

4.5.1.1 For chemical composition. Samples for chemical analysis shall be removed from each plate selected for ballistic testing.

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4.5.1.2 For mechanical properties tests.

(a) Rolled plate. A tension test specimen for first article tests shall be removed from the same plate material that has been selected for the ballistic test. The location and type specimen shall be as specified in 4.6.2(a).

(b) Forged material. One longitudinal and one transverse tension test specimen shall be removed from each forging submitted for ballistic test. Location and type of specimens shall be as specified in 4.6.2(b).

4.5.1.3 Stress corrosion tests. Stress corrosion test specimens shall be removed from the same plate or forging selected for the ballistic test. The type of specimens shall be as specified in 4.7.3. Location of specimens from forged material shall be as specified in the contract or order (see 6.2).

4.5.1.4 Ballistic tests.

(a) Rolled plate. Two plates, 12 by 36 inches by the ordered thickness, shall be submitted for ballistic testing. The orientation of these plates with respect to the rolling direction shall be at the option of the manufacturer. The ballistic test may be waived at the discretion of the procuring activity.

(b) Forged shapes. For each configuration that the manufacturer has under contract, three or more forgings shall be submitted for ballistic tests. Prior to submission of the forgings, the tester and the contractor will discuss, using a drawing of the forgings, the ballistic test conditions; namely, forging thickness, impact area and projectile type. The number of forgings, or forging sections needed for first article ballistic acceptance testing will be determined. Modifications, cutting, welding and bracketing will be determined as needed for test mounting.

4.5.2 For acceptance tests.

4.5.2.1 Chemical composition.

(a) For ladle analysis. At least one sample shall be taken from the molten metal representing one group of ingots poured as a unit from the same source of molten metal. Complete ingot analysis records shall be available to the Government at the ingot producer's facility.

(b) For check analysis. When sampling has not been made in accordance with 4.5.2.1(a), one sample shall be taken for each 4,000 pounds or less in each lot in accordance with applicable methods specified in ASTM E34 or ASTM 716.

4.5.2.2 Mechanical properties.

(a) Rolled plate. From each lot samples for tension tests shall be selected in accordance with table IV. Each sample shall be selected from a different plate in the lot. Should a lot consist of only one plate, only one tension test sample shall be taken. One tension test specimen shall be made from each sample.

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(b) Forged shapes. One forging for tension test shall be selected from each lot. One longitudinal and one transverse tension test specimen shall be removed from the forging. If size permits, this may be the same forging that is used for ballistic testing.

TABLE IV. Number of tension test samples.

Lot size, pounds	Minimum number of samples
To 8,000, incl.	2
8,001-12,000, incl.	3
12,001-20,000, incl.	4
Over 20,000, incl.	5

4.5.2.3 Stress corrosion tests (for surveillance testing only).

(a) Rolled plate. One plate, approximately 12 inches by 12 inches by the thickness (in inches) representing the lot, shall be selected for stress corrosion tests. At the discretion of the procuring activity, reduced sampling may be instituted (not every lot need be sampled).

(b) Forged shapes. One or more forged shapes (enough to provide the prescribed number of specimens) shall be selected for stress corrosion tests. Reduced sampling shall be at the discretion of the procuring activity.

4.5.2.4 Ballistic testing.

(a) Rolled plate. One plate approximately 12 inches by 36 inches by the thickness representing the lot shall be selected for ballistic testing. The orientation of this plate with respect to the rolling direction shall be at the option of the manufacturer.

(b) Forged shapes. From each lot, one forging shall be submitted for ballistic testing. (See 4.5.1.4[b] for minimal area requirements.)

4.6 Test specimens.

4.6.1 Chemical analysis. Samples for chemical analysis shall be prepared in accordance with the applicable methods specified in ASTM E34 or ASTM 716.

4.6.2 Mechanical properties.

(a) Rolled plate. Tension test specimens shall conform to the dimensions of the 0.500 inch diameter round specimen of ASTM B557. They shall be taken in the long transverse direction. From plate in nominal thickness 0.75 to 1.50 inches inclusive, tension test specimens shall be taken with the axis midway between the two plate surfaces; from plate in nominal thickness greater than 1.50 inches, the axis of the tension test specimen shall be three-fourths of the distance from one surface to the other (see 6.2).

(b) Forged shapes. Tension test specimens shall conform to the dimensions of the 0.500 inch diameter round specimen of ASTM B557. If the forging dimensions prevent the use of the 0.500 inch diameter specimen, the

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largest of the round subsize proportional specimens shown in ASTM B557 which can be obtained shall be used. Longitudinal tension test specimens shall be taken with the long axis parallel to the direction of maximum metal flow. Transverse specimens shall be taken with the long axis perpendicular to the direction of maximum metal flow. Actual location shall be as specified on the drawing or in the contract or order (see 6.2).

4.6.3 Stress corrosion. The stress corrosion specimens shall be stressed at 75% of the specified yield strength.

4.7 Test procedures.

4.7.1 Chemical analysis. Chemical analysis shall be conducted in accordance with the applicable methods specified in ASTM E34 or E227, E607, or E1251 of ASTM E716. The analysis shall comply with the declared composition established in accordance with the requirements of table I (see 3.2).

4.7.2 Mechanical properties. The mechanical properties shall be determined in accordance with ASTM B557.

4.7.3 Capability requirements for resistance to stress corrosion. When stress corrosion testing is specified, the test shall be conducted in accordance with ASTM G47. The location of the specimens taken from die forgings shall be specified in the forging drawing.

4.7.4 Ballistic tests.

(a) Rolled plate. Prior to testing, the plates shall be temperature conditioned (see Appendix A, A.4.2). Plates shall be tested using projectiles chosen in accordance with the thickness ranges shown in table V. Plate thickness as determined by the ballistic test agency shall be used to determine the required ballistic limit for that plate. Individual thickness measurements are to be read from a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 inch and the average of these readings reported to the nearest 0.005 inch. Measurements shall be made on the intended impact area.

TABLE V. Acceptance ballistic test plates or forgings.

Ordered or nominal design thickness specified on drawings, inches	Projectile to be used
0.700-0.950	Cal. .50 FS
0.951-1.550	20 mm FS
1.551-3.000	Cal. .50 AP, M2
3.001-4.000	14.5 mm API, BS41

(b) Forged shapes. Prior to testing, the forgings shall be temperature conditioned (see Appendix A, A.4.2). Ballistic test forgings shall be tested at locations specified (see 6.2). Forgings shall be tested using projectiles chosen in accordance with thickness ranges shown in table A-IV. Unless otherwise stated, required minimum ballistic limit protection shall be the measured thickness of the forging.

4.7.5 Forging thickness measurements. The average thickness of the forging shall be the mean of at least 4 thickness measurements read to the

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nearest 0.01 inch. These measurements shall be made in regions of uniform thickness as close as possible to the intended point of impact of projectiles within the area required to provide ballistic protection as shown on the drawing.

4.8 Examination.

4.8.1 Dimensions.

4.8.1.1 Rolled plate. Plates within the lot shall be measured to determine compliance with the requirements for dimensions and tolerances (see 3.7) in accordance with sampling procedures approved by the procuring activity.

4.8.1.2 Forged shapes. From each acceptance lot, representative forgings shall be selected in accordance with table VI and examined to determine compliance with the requirements of 3.7.

TABLE VI. Sampling for dimensional examination.

Number of forgings in the lot	Number of sample forgings
Up to 50, incl.	5
51 to 130, incl.	10
131 to 300, incl.	15

4.8.2 Visual. Each plate or forging shall be examined to determine compliance with the identification marking (see 3.8) and workmanship (see 3.10) requirements.

4.8.3 Preparation for shipment. Prior to shipment, examination shall be made to determine compliance with section 5.

4.9 Rejection and retest.

4.9.1 Rejection. Unless otherwise specified in the contract or order (see 6.2), failure of the first article test plates or forgings to meet the requirements of 4.3.1 shall be cause for rejection of the process, failure of the acceptance samples to meet the requirements of 4.8 shall be cause for rejection of the lot (see 4.9.2).

4.9.2 Retest. Unless specific retest procedure is specified in the contract or order (see 6.2), samples shall be submitted for each failed sample in accordance with paragraphs 4.10.1, 4.10.2 or 4.10.3. Failure of either of the retest samples (plates) shall be cause for rejection of the material. First article retests shall not be permitted until the supplier has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Samples submitted for retest shall be marked R1, R2 and R3, if applicable, in addition to the requirements of 3.8.

4.10 Acceptance.

4.10.1 Stress corrosion (first article) testing only. If the manufacturer so desires, a retest shall be performed using three additional specimens. No cracking shall occur in 10 days. If the retest also fails, the manufacturer may elect to resubmit the lot after retreatment of the entire lot. After retreatment, the lot must conform to the requirements outlined in 4.7.3.

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4.10.2 Ballistic. Rejection and retest of ballistic test plates and forged shapes shall be in accordance with Appendix A (see A.4.4).

4.10.3 Reduced testing. At the discretion of the procuring activity, the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product, meeting the testing requirements is being produced and provided the manufacturer agrees to maintain the same manufacturing procedures.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The material specified herein is military unique. This material has minimum ballistic limit protection requirements as specified in tables A-I through A-IV. The material will be used in military vehicles to protect the occupants against small arms fire and shrapnel.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of material (see 1.3).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and A.2.1).
(d) When first article is required (see 3.1).
(e) Mechanical properties, if different from table II (see 3.3).
(f) Whether stress-corrosion testing is to be specified (see 3.5).
(g) Drawings, dimensions and tolerances (see 3.7 and subparagraphs).
(h) Ballistic test report (see 3.9).
(i) Destination of test samples (see 4.2).
(j) Location and orientation of stress corrosion samples from forged material (see 4.5.1.3 and 4.7.3).
(k) Location of tension test specimens (see 4.6.2[a] and [b]).
(l) Location for determination of ballistic limits (see 4.7.4[b]).
(m) Whether a retest procedure is to be specified (see 4.9.2).
(n) Packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Contractor. The term "contractor" is defined as the organization having a direct contract with the procuring activity.

6.3.2 Contracting officer. The term "contracting officer" means the person executing a contract on behalf of the Government and any other officer

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or civilian employee who is properly designated contracting officer; and the term includes, except as otherwise provided, the authorized representative of a contracting officer acting within the limits of his authority.

6.3.3 Manufacturer. The term “manufacturer” is defined as the organization actually performing the operations covered by this specification.

6.3.4 Procuring activity. The term "procuring activity" is that activity of the Government which actually initiates the request for procurement and maintains the records of the procurement.

6.4 Mechanical properties and ballistic requirements. The minimum mechanical properties specified (see 3.3) may not assure aluminum armor material meeting the specified ballistic requirements (see 3.4).

6.5 Alloy number. Aluminum Association alloy number 2219 is the same material as listed under current UNS designation as A92219.

6.6 Subject term (key word) listing.

Ballistic tests
Military vehicles
Stress corrosion

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

BALLISTIC TESTING OF ALUMINUM ALLOY ARMOR,
2219, ROLLED PLATE AND DIE FORGED SHAPES

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor, 2219, rolled plate and die forged shapes, when tested in accordance with the provisions of this specification. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification and the information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-662 - V_{50} Ballistic Test for Armor

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Automated Printing Service (DAPS), Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

A.3 REQUIREMENTS

A.3.1 Resistance to penetration. The minimum required V_{50} ballistic limit shall be in accordance with the values shown in tables A-I through A-IV.

A.4 VERIFICATION

A.4.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V_{50} Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.4.2 Temperature conditioning. Prior to the test, the test item(s) will be temperature conditioned at least eight hours. Thermostatic control will be such that the average temperature of the item during the test shall be 72° $\pm$ 15°F (22° $\pm$ 8°C).

A.4.3 Ballistic limit protection, BL(P).

A.4.3.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration

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velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) will be reported.

A.4.3.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data will be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the forged thickness, the ballistic limit will be computed on the basis of 4 or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing will continue until a 10-round ballistic limit has been attained using the smallest practical velocity spread. Ten round ballistic limits shall be reported as such in a ballistic test report as specified in MIL-STD-662.

A.4.3.3 Rejection of large velocity gap in borderline cases. If the ballistic limit which has been attained is within ± 10 fps the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, will be fired to reduce the gap to 25 fps or less. The ballistic limit will then be recomputed using the above criteria. The recomputed BL(P) will be reported as the BL(P) of the plate. In borderline cases, a reduction of the gap between the high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P).

A.4.4 Rejection and retest of ballistic plates.

A.4.4.1 First article tests (rejection). Unless otherwise specified in the contract or order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

A.4.4.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates shall be submitted for first article testing and both tests must pass; otherwise, the armor material shall be rejected.

A.4.4.3 Acceptance tests (rejection). Unless otherwise specified in the contract or order, failure of a test plate to meet the ballistic requirements indicates failure of the lot; however, the final decision will depend on the outcome of retests, if submitted.

A.4.4.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure, may submit at his own expense two additional test plates from the same lot for ballistic retest. If either of these plates fail the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fail, the lot shall be permanently rejected.

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APPENDIX A

A.4.5 Ownership of ballistic test samples.

A.4.5.1 First article and quality conformance test material complying to ballistic testing. First article and quality conformance test material that complies with the requirements of this specification are considered as part of the lot of material that they represent, and ownership of them passes to the Government with acceptance of that lot. Test material that fails to comply with the requirements of this specification are considered as part of the lot they represent and remain property of the producer as does the rejectable lot which they represent.

A.4.5.2 First article and conformance test material failing ballistic testing. At the time of shipment of the first article and conformance test material, the Government test activity should be informed by the manufacturer, in writing, as to the disposition of the test material which fails the test. If instructions for disposal of the tested material are not received after 15 days, the Government test activity will exercise the prerogative of disposing of the material.

A.6 DEFINITIONS

A.6.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

A.6.2 Fair impact. An impact may be considered fair when an unyawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

A.6.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

A.6.4 Partial penetration, (PP). Any impact which is not a complete penetration may be considered a partial penetration.

A.6.5 Witness plate. A thin sheet located behind and parallel to the ballistic test sample which is used to detect penetrating projectiles or spall.

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APPENDIX A

TABLE A-I. Minimum required ballistic limits (protection at 0° obliquity).

Caliber .50 Fragment Simulating Projectiles

Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps
0.700[1/]	1666	0.800	2054	0.900	2530
.705	1683	.805	2076	.905	2556
.710	1701	.810	2097	.910	2583
.715	1719	.815	2119	.915	2610
.720	1737	.820	2142	.920	2637
.725	1755	.825	2164	.925	2665
.730	1774	.830	2187	.930	2693
.735	1793	.835	2210	.935	2721
.740	1811	.840	2233	.940	2749
.745	1831	.845	2256	.945	2778
.750	1850	.850	2280	.950[2/]	2807
.755	1869	.855	2304	.955	2836
.760	1889	.860	2328	.960	2866
.765	1909	.865	2352	.965	2895
.770	1929	.870	2377	.970	2926
.775	1949	.875	2402	.975	2956
.780	1970	.880	2427	.980	2987
.785	1990	.885	2452	.985	3018
.790	2012	.890	2478	.990	3049
.795	2033	.895	2504	.995	3081

1/ Specification ballistic test requirements begin at ordered thickness 0.700 inch.

2/ Specification ballistic test requirements end at ordered thickness 0.950 inch.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits (protection at 0° obliquity).

20mm FS Projectile

Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps
0.900	1210	1.175	1735	1.450	2500
.905	1219	1.180	1746	1.455	2515
.910	1228	1.185	1757	1.460	2530
.915	1237	1.190	1768	1.465	2545
.920	1246	1.195	1779	1.470	2560
.925	1255	1.200	1790	1.475	2575
.930	1264	1.205	1802	1.480	2590
.935	1273	1.210	1814	1.485	2605
.940	1282	1.215	1826	1.490	2620
.945	1291	1.220	1838	1.495	2635
.950	1300	1.225	1850	1.500	2650
.955	1309	1.230	1862	1.505	2665
.960	1318	1.235	1874	1.510	2680
.965	1327	1.240	1886	1.515	2695
.970	1336	1.245	1898	1.520	2710
.975	1345	1.250	1910	1.525	2725
.980	1354	1.255	1924	1.530	2740
.985	1363	1.260	1938	1.535	2755
.990	1372	1.265	1952	1.540	2770
.995	1381	1.270	1966	1.545	2785
1.000	1390	1.275	1980	1.550	2800
1.005	1399	1.280	1994	1.555	2815
1.010	1408	1.285	2008	1.560	2830
1.015	1417	1.290	2022	1.565	2845
1.020	1426	1.295	2036	1.570	2860
1.025	1435	1.300	2050	1.575	2875
1.030	1444	1.305	2065	1.580	2890
1.035	1453	1.310	2080	1.585	2905
1.040	1462	1.315	2095	1.590	2920
1.045	1471	1.320	2110	1.595	2935
1.050	1480	1.325	2125	1.600	2950
1.055	1490	1.330	2140	1.605	2965
1.060	1500	1.335	2155	1.610	2980
1.065	1510	1.340	2170	1.615	2995
1.070	1520	1.345	2185	1.620	3010
1.075	1530	1.350	2200	1.625	3025
1.080	1540	1.355	2215	1.630	3040
1.085	1550	1.360	2230	1.635	3055
1.090	1560	1.365	2245	1.640	3070
1.095	1570	1.370	2260	1.645	3085
1.100	1580	1.375	2275	1.650	3100
1.105	1590	1.380	2290	1.655	3115
1.110	1600	1.385	2305	1.660	3130
1.115	1610	1.390	2320	1.665	3145
1.120	1620	1.395	2335	1.670	3160
1.125	1630	1.400	2350	1.675	3175
1.130	1640	1.405	2365	1.680	3190
1.135	1650	1.410	2380	1.685	3205
1.140	1660	1.415	2395	1.690	3220
1.145	1670	1.420	2410	1.695	3235
1.150	1680	1.425	2425	1.700	3250
1.155	1691	1.430	2440		
1.160	1702	1.435	2455		
1.165	1713	1.440	2470		
1.170	1724	1.445	2485		

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APPENDIX A

TABLE A-III. Minimum required ballistic limits (protection at 0° obliquity).

Caliber .50 AP, M2 projectile

Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps
1.500	1924	1.745	2106	1.995	2277
1.505	1928	1.750	2110	2.000	2280
1.510	1932	1.755	2113	2.005	2284
1.515	1936	1.760	2117	2.010	2287
1.520	1940	1.765	2120	2.015	2290
1.525	1943	1.770	2124	2.020	2294
1.530	1947	1.775	2127	2.025	2297
1.535	1951	1.780	2131	2.030	2300
1.540	1955	1.785	2135	2.035	2303
1.545	1959	1.790	2138	2.040	2307
1.550	1963	1.795	2142	2.045	2310
* 1.551	1963	1.800	2145	2.050	2313
1.555	1966	1.805	2149	2.055	2316
1.560	1970	1.810	2152	2.060	2320
1.565	1974	1.815	2155	2.065	2323
1.570	1978	1.820	2159	2.070	2326
1.575	1982	1.825	2162	2.075	2329
1.580	1985	1.830	2166	2.080	2332
1.585	1989	1.835	2169	2.085	2336
1.590	1993	1.840	2173	2.090	2339
1.595	1997	1.845	2176	2.095	2342
1.600	2000	1.850	2180	2.100	2345
1.605	2004	1.855	2183	2.105	2348
1.610	2008	1.860	2187	2.110	2352
1.615	2012	1.865	2190	2.115	2355
1.620	2015	1.870	2193	2.120	2358
1.625	2019	1.875	2197	2.125	2361
1.630	2023	1.880	2200	2.130	2364
1.635	2026	1.885	2204	2.135	2368
1.640	2030	1.890	2207	2.140	2371
1.645	2034	1.895	2210	2.145	2374
1.650	2038	1.900	2214	2.150	2377
1.655	2041	1.905	2217	2.155	2380
1.660	2045	1.910	2221	2.160	2383
1.665	2049	1.915	2224	2.165	2386
1.670	2052	1.920	2227	2.170	2390
1.675	2056	1.925	2231	2.175	2393
1.680	2059	1.930	2234	2.180	2396
1.685	2063	1.935	2237	2.185	2399
1.690	2067	1.940	2241	2.190	2402
1.695	2070	1.945	2244	2.195	2405
1.700	2074	1.950	2247	2.200	2408
1.705	2078	1.955	2251	2.205	2411
1.710	2081	1.960	2254	2.210	2415
1.715	2085	1.965	2257	2.215	2418
1.720	2088	1.970	2261	2.220	2421
1.725	2092	1.975	2264	2.225	2424
1.730	2096	1.980	2267	2.230	2427
1.735	2099	1.985	2271	2.235	2430
1.740	2103	1.990	2274	2.240	2433

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APPENDIX A

TABLE A-III. Minimum required ballistic limits (protection at 0° obliquity) - Continued.

Caliber .50 AP, M2 projectile

Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps
2.245	2436	2.505	2591	2.765	2737
2.250	2439	2.510	2594	2.770	2740
2.255	2442	2.515	2597	2.775	2743
2.260	2445	2.520	2600	2.780	2746
2.265	2448	2.525	2603	2.785	2748
2.270	2451	2.530	2606	2.790	2751
2.275	2455	2.535	2608	2.795	2754
2.280	2458	2.540	2611	2.800	2756
2.285	2461	2.545	2614	2.805	2759
2.290	2464	2.550	2617	2.810	2762
2.295	2467	2.555	2620	2.815	2765
2.300	2470	2.560	2623	2.820	2767
2.305	2473	2.565	2626	2.825	2770
2.310	2476	2.570	2628	2.830	2773
2.315	2479	2.575	2631	2.835	2775
2.320	2482	2.580	2634	2.840	2778
2.325	2485	2.585	2637	2.845	2781
2.330	2488	2.590	2640	2.850	2784
2.335	2491	2.595	2643	2.855	2786
2.340	2494	2.600	2646	2.860	2789
2.345	2497	2.605	2648	2.865	2792
2.350	2500	2.610	2651	2.870	2794
2.355	2503	2.615	2654	2.875	2797
2.360	2506	2.620	2657	2.880	2800
2.365	2509	2.625	2660	2.885	2802
2.370	2512	2.630	2662	2.890	2805
2.375	2515	2.635	2665	2.895	2808
2.380	2518	2.640	2668	2.900	2810
2.385	2521	2.645	2671	2.905	2813
2.390	2524	2.650	2674	2.910	2816
2.395	2527	2.655	2676	2.915	2818
2.400	2530	2.660	2679	2.920	2821
2.405	2533	2.665	2682	2.925	2824
2.410	2536	2.670	2685	2.930	2826
2.415	2539	2.675	2688	2.935	2829
2.420	2542	2.680	2690	2.940	2832
2.425	2544	2.685	2693	2.945	2834
2.430	2547	2.690	2696	2.950	2837
2.435	2550	2.695	2699	2.955	2839
2.440	2553	2.700	2702	2.960	2842
2.445	2556	2.705	2704	2.965	2845
2.450	2559	2.710	2707	2.970	2847
2.455	2562	2.715	2710	2.975	2850
2.460	2565	2.720	2713	2.980	2853
2.465	2568	2.725	2715	2.985	2855
2.470	2571	2.730	2718	2.990	2858
2.475	2574	2.735	2721	2.995	2861
2.480	2577	2.740	2724	** 3.000	2863
2.485	2580	2.745	2726		
2.490	2582	2.750	2729		
2.495	2585	2.755	2732		
2.500	2588	2.760	2735		

* Specification requirements begin for this ordered thickness.
** Specification requirements end for this ordered thickness.
Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits (protection at 0° obliquity).

14.5 mm API, BS 41 projectile

Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps
2.900	2655	3.075	2774	3.250	2890
2.905	2659	3.080	2778	3.255	2894
2.910	2662	3.085	2781	3.260	2897
2.915	2666	3.090	2784	3.265	2900
2.920	2669	3.095	2788	3.270	2904
2.925	2673	3.100	2791	3.275	2907
2.930	2676	3.105	2794	3.280	2910
2.935	2680	3.110	2797	3.285	2914
2.940	2683	3.115	2801	3.290	2917
2.945	2687	3.120	2804	3.295	2920
2.950	2690	3.125	2807	3.300	2924
2.955	2694	3.130	2811	3.305	2927
2.960	2697	3.135	2814	3.310	2930
2.965	2701	3.140	2817	3.315	2934
2.970	2704	3.145	2821	3.320	2937
2.975	2708	3.150	2824	3.325	2940
2.980	2711	3.155	2827	3.330	2944
2.985	2715	3.160	2830	3.335	2947
2.990	2718	3.165	2834	3.340	2950
2.995	2722	3.170	2837	3.345	2954
3.000	2725	3.175	2840	3.350	2957
3.005	2728	3.180	2844	3.355	2960
3.010	2731	3.185	2847	3.360	2964
3.015	2735	3.190	2850	3.365	2967
3.020	2738	3.195	2854	3.370	2970
3.025	2741	3.200	2857	3.375	2974
3.030	2745	3.205	2860	3.380	2977
3.035	2748	3.210	2864	3.385	2980
3.040	2751	3.215	2867	3.390	2984
3.045	2755	3.220	2870	3.395	2987
3.050	2758	3.225	2874	3.400	2990
3.055	2761	3.230	2877	3.405	2994
3.060	2764	3.235	2880	3.410	2997
3.065	2768	3.240	2884	3.415	3000
3.070	2771	3.245	2887	3.420	3004
				3.425	3007

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Army - AT, TE
DLA - IS

Preparing activity:
Army - MR

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-46118E (MR)	**2. DOCUMENT DATE (YYMMDD)** 98-08-26

DOCUMENT TITLE ALUMINUM ALLOY ARMOR, 2219, ROLLED PLATE AND DIE FORGED SHAPES

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US ARMY RESEARCH LABORATORY	b. TELEPHONE *Include Area Code)* (1) Commercial (410) 306-0725 (2) AUTOVON 458-0725
c. ADDRESS *(Include Zip Code)Commander* WEAPONS & MATERIAL RESEARCH DIRECTORATE ATTN; AMSRL-WM-M ABERDEEN PROVING GROUND, MD 21005-5069	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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